NAME - KRISH CHANDRASHEKHAR PACHODE ROLL NO. - B-18 BATCH - AIDS (SY-B)

AIM - Implement and analyze various probability distributions in Python, including Binomial, Poisson, and Normal distributions.

THEORY -

1. What is the difference between Binomial, Poisson, and Normal distributions?

- **Binomial Distribution:** Used for a fixed number of independent trials with two outcomes (success or failure). Example: Tossing a coin multiple times.
- **Poisson Distribution:** Used to model the number of events occurring in a fixed interval of time or space. Example: Number of calls received in a call center per hour.
- **Normal Distribution:** A continuous probability distribution that is symmetric and bell-shaped. Example: Heights or marks of students.

 Binomial is discrete with fixed trials, Poisson is discrete for rare events, and Normal is continuous.

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2. What is the real-world significance of the Poisson distribution?

Poisson distribution is useful for predicting the number of events occurring in a fixed interval.

Applications:

- Number of customers arriving at a shop
- Number of emails received per hour
- Number of accidents in a day

 It helps in modeling rare or random events in real life.

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3. What is the Central Limit Theorem and how does it relate to Normal distribution?

The **Central Limit Theorem (CLT)** states that the distribution of sample means approaches a normal distribution as the sample size increases, regardless of the original data distribution.

 This means even if data is not normally distributed, the sample mean will follow a normal distribution for large samples.

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4. Why is Normal distribution widely used in machine learning and statistics?

- It is simple and mathematically convenient
- Many real-world data follow normal distribution
- Used in statistical tests and probability models
- Helps in making predictions and assumptions

 It forms the basis of many machine learning algorithms.

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5. What is skewness in distributions and which distributions exhibit it?

Skewness refers to the asymmetry in a distribution.

- **Positive Skew:** Tail on the right side
- **Negative Skew:** Tail on the left side

Examples:

- Poisson distribution can be skewed
- Binomial can be skewed depending on probability
- Normal distribution is not skewed (perfectly symmetric)

```
from google.colab import files
uploaded = files.upload()
```

```
<IPython.core.display.HTML object>

Saving Teen_Mental_Health_Dataset.csv to
Teen_Mental_Health_Dataset.csv
```

```
!pip install scipy
```

```
Requirement already satisfied: scipy in
/usr/local/lib/python3.12/dist-packages (1.16.3)
Requirement already satisfied: numpy<2.6,>=1.25.2 in
/usr/local/lib/python3.12/dist-packages (from scipy) (2.0.2)
```

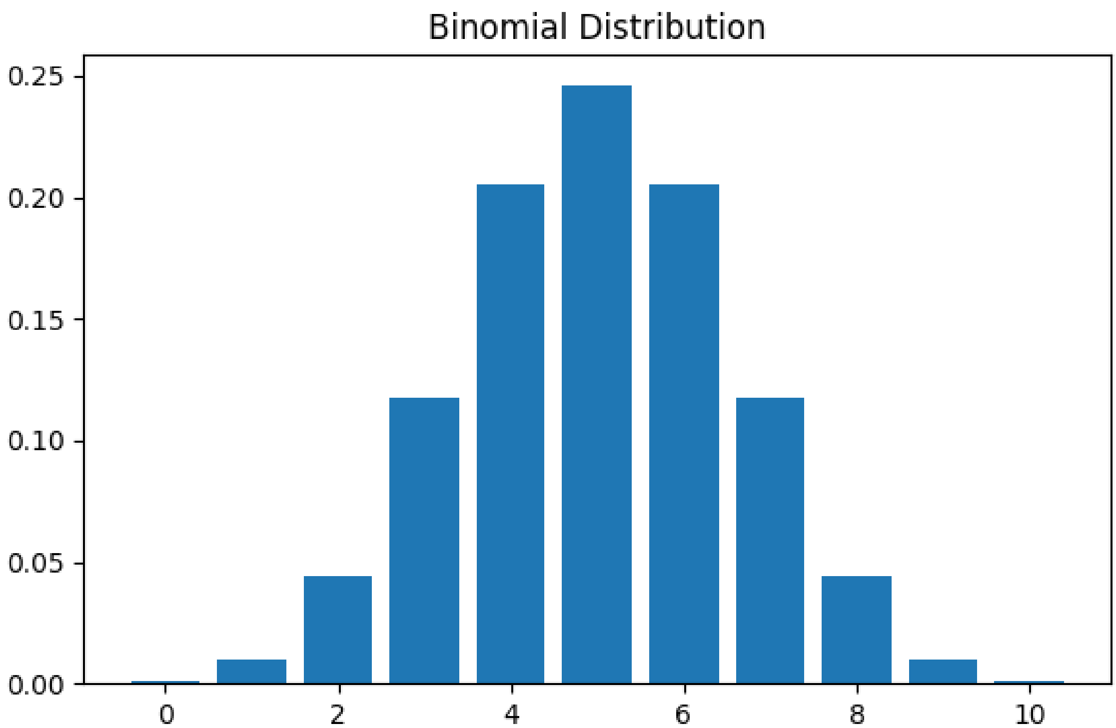
```
# Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import binom, poisson, norm

# Load dataset (use uploaded file name)
df = pd.read_csv("Teen_Mental_Health_Dataset.csv")
```

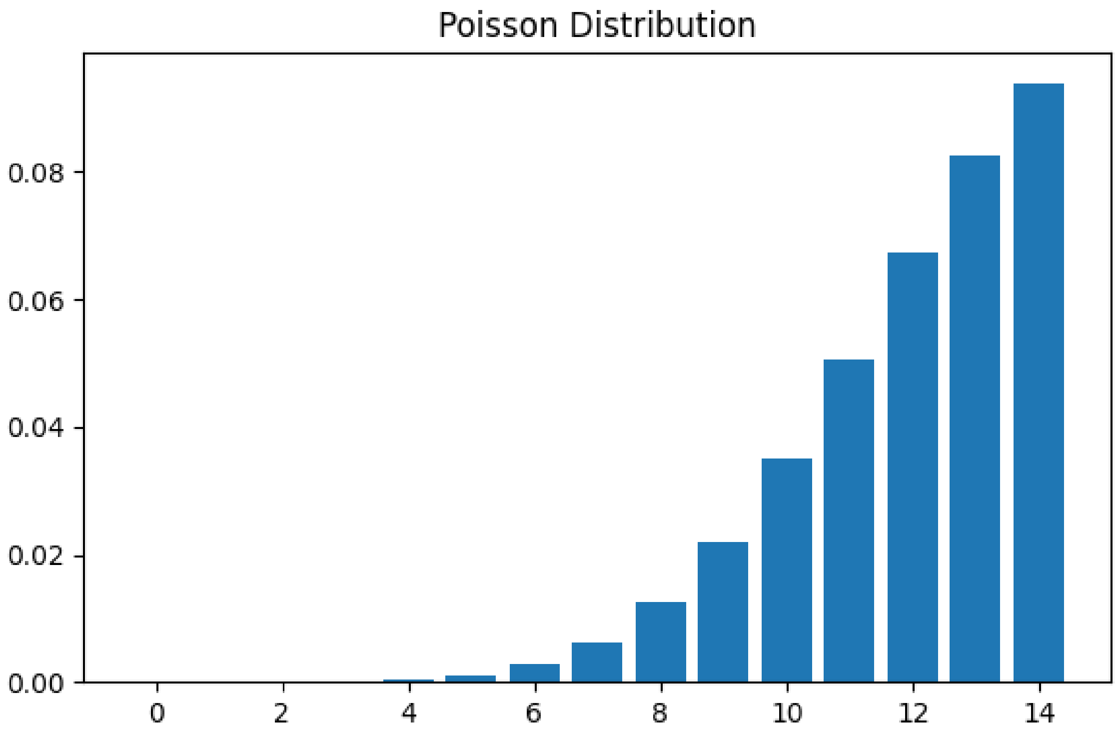
```python
print("Dataset Loaded")
print(df.head())

# Select numeric column
numeric_col = df.select_dtypes(include=['number']).columns[0]
data = df[numeric_col].dropna()

print("Using Column:", numeric_col)

# --------------------------------
# BINOMIAL DISTRIBUTION
# --------------------------------
n = 10
p = 0.5
x = np.arange(0, n+1)

binomial_pmf = binom.pmf(x, n, p)

plt.figure(figsize=(6,4))
plt.bar(x, binomial_pmf)
plt.title("Binomial Distribution")
plt.tight_layout()
plt.show()

# Save CSV
pd.DataFrame({"x": x, "binomial":
binomial_pmf}).to_csv("binomial.csv", index=False)

# --------------------------------
# POISSON DISTRIBUTION
# --------------------------------
lam = data.mean()
x = np.arange(0, 15)

poisson_pmf = poisson.pmf(x, lam)

plt.figure(figsize=(6,4))
plt.bar(x, poisson_pmf)
plt.title("Poisson Distribution")
plt.tight_layout()
plt.show()

# Save CSV
pd.DataFrame({"x": x, "poisson": poisson_pmf}).to_csv("poisson.csv",
index=False)

# --------------------------------
# NORMAL DISTRIBUTION
# --------------------------------
```

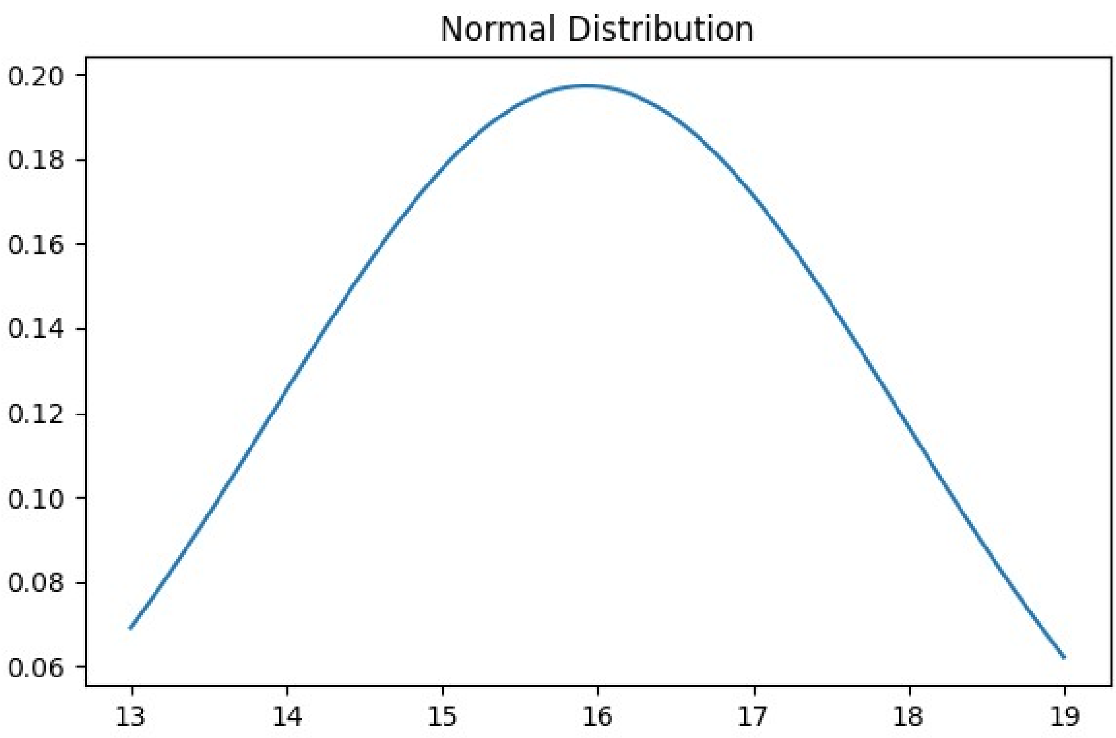
```
mean = data.mean()
std = data.std()

x = np.linspace(data.min(), data.max(), 100)
normal_pdf = norm.pdf(x, mean, std)

plt.figure(figsize=(6,4))
plt.plot(x, normal_pdf)
plt.title("Normal Distribution")
plt.tight_layout()
plt.show()

# Save CSV
pd.DataFrame({"x": x, "normal": normal_pdf}).to_csv("normal.csv",
index=False)

print("All CSV files created successfully!")
```

```
Dataset Loaded
   age  gender  daily_social_media_hours platform_usage
sleep_hours  \
0   14    male                       7.9      Instagram          7.4

1   19  female                       1.9         TikTok          8.0

2   17  female                       1.3      Instagram          7.6

3   15    male                       7.4         TikTok          6.9

4   15  female                       4.7           Both          4.9


   screen_time_before_sleep  academic_performance
physical_activity  \
0                       2.9                  3.01                  1.5

1                       2.9                  3.22                  0.8

2                       0.5                  3.92                  0.0

3                       1.6                  3.48                  0.8

4                       3.0                  2.37                  1.4


  social_interaction_level  stress_level  anxiety_level
addiction_level  \
0                      low             2              2
1
1                     high             8              1
10
```

```
2                       high             2              4
2
3                     medium             1              7
9
4                     medium             3              5
2

   depression_label
0                 0
1                 0
2                 0
3                 0
4                 0
Using Column: age
```








```
All CSV files created successfully!
```

Conclusion –

In this experiment, we implemented Binomial, Poisson, and Normal distributions using Python. We visualized these distributions to understand their behavior and applications. These probability distributions are widely used in statistics, data analysis, and machine learning for modeling real-world problems.